MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Linear and affine maps

Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \to \mathbb{R}^m.$$

A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition: Matrix map Given an $m \times n$ matrix A, the $\boxed{\qquad\qquad}$ induced by A is the function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition: Linear map A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is $\boxed{\qquad\qquad}$ if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}),$$
$$T(c\mathbf{v}) = cT(\mathbf{v})$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c.

Theorem: Matrix maps are linear If A is an $m \times n$ matrix, then the function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ is linear.

Proof Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \qquad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Theorem: Every linear map has a matrix Every linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Activity: Images of basis vectors determine the matrix

Suppose a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (-1, 3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

Warning We use "linear map" and "linear transformation" interchangeably; "transformation" is often used when emphasizing geometry.

Definition: Affine map A function of the form

$$\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$$

is called ________. It is built from a linear map followed by a translation.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

Example: An affine layer A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.